

CENG381 Stochastic Processes

Classifying States — Practice 1

Q1 (30 points). Consider the Markov chain on $\{1, 2, 3, 4, 5, 6\}$ with:

$$\begin{aligned}P(1, 2) &= 1/2, & P(1, 3) &= 1/2 \\P(2, 1) &= 1 \\P(3, 4) &= 1/3, & P(3, 5) &= 1/3, & P(3, 6) &= 1/3 \\P(4, 3) &= 1 \\P(5, 5) &= 1 \quad (\text{absorbing}) \\P(6, 6) &= 1 \quad (\text{absorbing})\end{aligned}$$

- Identify all communicating classes. State which classes are closed (no escape) and which are not.
 - Classify each state as recurrent or transient. Recall: a state is recurrent if $P(\text{ever return} \mid \text{start there}) = 1$; transient otherwise.
 - Starting from state 3, what is the probability that the chain is eventually absorbed in state 5? (Hint: write a balance equation for $h = P(\text{reach 5} \mid \text{start 3})$.)
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Q2 (35 points). A production line is always in one of three states: Working (W), Under Maintenance (M), or Broken (B). The daily transition matrix is:

	W	M	B
W	[3/4	1/4	0]
M	[1/2	0	1/2]
B	[0	1	0]

- Is this chain irreducible? What does this imply about the classification of each state?
 - Find the stationary distribution π by solving $\pi \cdot P = \pi$ and $\pi_W + \pi_M + \pi_B = 1$.
 - The expected return time to state W equals $1/\pi_W$. Compute it. Interpret: on average, how many days pass between consecutive days of full operation?
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Q3 (35 points). A cereal company places one of $n = 4$ different trading cards in each box. Each card type is equally likely. You buy boxes one at a time until you collect all 4 distinct cards. Let T be the total number of boxes purchased.

- Model this as a Markov chain where the state is the number of distinct cards collected so far. Write down the states, identify which is absorbing, and give the transition probabilities $P(k, k+1)$

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and $P(k, k)$ for each transient state k .

b) The expected time to go from k distinct cards to $k+1$ distinct cards is $4/(4-k)$. Use this to compute $E[T]$ starting from 0 cards. Verify using the coupon-collector formula $E[T] = n \cdot (1 + 1/2 + 1/3 + 1/n)$.

c) Having already collected 2 distinct cards, find the probability that you have all 4 cards after exactly 2 more purchases.